Impact of Ring Current Asymmetry on Prompt Penetration Electric Fields during march 17^{th} 2015 Geomagnetic Storm at mid-low latitudes

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Abstract

This study analyzes the impact of the asymmetric ring current and Prompt Penetration Electric Fields on the local geomagnetic response, using data from four mid-to-low latitude observatories separated by approximately 180° in longitude, covering different local time sectors. We focus on the geomagnetic storm of 17 March 2015. Using Fourier analysis, we reconstruct the magnetic signatures related to PPEF. Our results demonstrate that the local geomagnetic response is most significantly affected in the dawn and dusk sectors, where the contribution of the partial ring current and the Field Aligned Currents to the local magnetic record is most pronounced. At the same time, our findings suggest that the asymmetric responses should not be generally considered, since the contribution will not always be homogeneous through different periods and positions.

Keywords: Regional Space weather, Magnetic fluctuations, Prompt-Penetration Electric Fields, asymmetric ring current

1 Introduction

A geomagnetic storm (GS) is a temporary disturbance of Earth's magnetosphere caused by the injection of energy and charged particles into the ring current system (RC), which leads to a temporal attenuation of the H component of the geomagnetic field intensity. This reduction is quantified by geomagnetic indices such as the DST index, which is sampled hourly, or the SYM-H index, which is sampled at minute intervals. These storms are typically triggered by solar phenomena, including coronal mass ejections (CME) or high-speed solar wind (HSSW) streams. Due to their potential impact on technological systems—such as disruptions to communications, navigation, and power grids—geomagnetic storms are among the most extensively studied phenomena in space weather research.

The main phase (MP) of a geomagnetic storm is the stage during which energy and charged particles from a CME or HSSW are transported into Earth's inner magnetosphere upon arrival. The magnetosphere is dynamically coupled with the high-latitude ionosphere [C.T Russell, 2016] through field-aligned currents (FAC). These are sheet current systems consisting of two regions. In Region 1 (R1), injection from the solar wind occurs on the dawn side into the ionosphere and exits on the dusk side. In Region 2 (R2), the current direction is opposite (upward from the ionosphere on the dawn side and downward on the dusk side), with coupling points located more equatorward.

In the magnetosphere, R2-FACs connect in the equatorial plane with a westward current flow distinct from the RC, known as the partial ring current (PRC). This current system is centered around the evening sector but exhibits a small, swift westward movement, as reported by Kamide and Fukushima [1971], Crooker and McPherron [1972], Fukushima and Kamide [1973]. As shown in Figure 1, from the perspective of the northern hemisphere, the PRC closure points with FACs involve downward current (FAC to PRC) on the eastern side and upward current (PRC to FAC) on the western afternoon side. This configuration results in a longitudinal asymmetry in the H-component decrease at low latitudes during the MP, contrasting with the symmetric effect of the ring current. Consequently, the geomagnetic

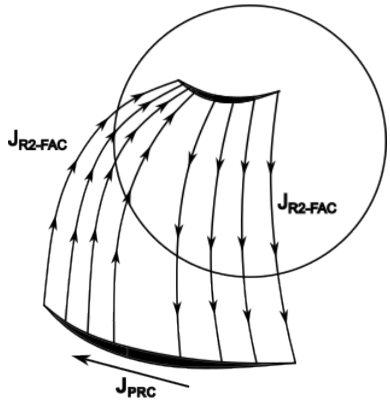


Figure 1: Simple model explaining the R2FAC-PRC circuit. From the equatorial plane perspective, notice that even when PRC is only connected with R2-FAC, there is a region of descending flux in the eatern side and an ascending flux in the western side. Adapted from Fukushima and Kamide [1973].

response is more complex than expected, as Dst and SYMH indices only account for the symmetric component.

According to Fukushima and Kamide [1973], FAC is the main source for the asymmetric response of what they call *asymmetric ring current* (ARC). In this ARC, the most negative H response is related with the upward closure circuit points with PRC, around 18-20 magnetic local time (MLT) sector during early MP stages.

In the high-latitude ionosphere, coupling with R1-FACs enables the propagation of interplanetary electric fields (IEF), which can in turn penetrate to lower ionospheric latitudes, giving rise to prompt penetration electric fields (PPEF). The concept of PPEF was first introduced by Nishida [1968a,b], who observed periodic magnetic fluctuations with periods ranging from 30 minutes to 4 hours, detected by groups of ground magnetometers located around the auroral oval and the dip equator. He reported that magnetometer pairs at similar longitudes—one in the auroral region and the other near the equator—exhibited nearly identical fluctuations, with a correlation coefficient of approximately 0.9.

Later, Vasyliunas [1970], Jaggi and Wolf [1973] developed the first model explaining the penetration of IEFs through R1-FAC. Under normal conditions, these fields are shielded by the action of R2-FAC. However, during periods of intense R1-FAC activity, a state known as *undershielding* occurs, allowing IEFs to propagate through the ionosphere until R2-FAC become intensified by PRC activity. This intensification can lead to an *overshielding* state, representing another imbalance condition. The alternating sequence of undershielding and overshielding gives rise to the fluctuations in PPEFs, which can be recorded by ground magnetometers. This fluctuation process can persist for several hours [Wei et al., 2015] until the reconnection process—and consequently the MP—ceases. As noted by Kelley et al. [1979], Wei et al. [2015], PPEF occurrence depends on both the southward orientation of the interplanetary magnetic field (IMF) B_Z component and sharp directional changes in B_Z.

As illustrated in Figure 2, when IEF—now acting as PPEF—are transmitted to the high-latitude ionosphere, they drive Hall and westward Pedersen currents. These PPEF fluctuations can propagate horizontally through the ionosphere via electromagnetic waves toward mid- and low-latitudes [Kikuchi and Araki, 1979, Wei et al., 2015]. Although PPEF amplitudes attenuate with decreasing latitude, they become intensified at equatorial latitudes due to enhanced conductivity in the equatorial ionosphere, giving rise to an equatorial westward Pedersen current. This Pedersen current is depicted in Figure 2 as a red line, with arrows indicating the flow direction at equatorial latitudes and current closure at adjacent latitudes near the terminators.

On the dayside at mid-to-low latitudes ($20 - 30°$), an eastward prompt penetration electric field (PPEF) induces an upward $\mathbf{E} \times \mathbf{B}$ drift, lifting plasma to higher altitudes where recombination is slower, often resulting in a positive ionospheric storm (characterized by an increase in total electron content, TEC). In contrast, storm-induced circulation from high to low latitudes transports molecular-

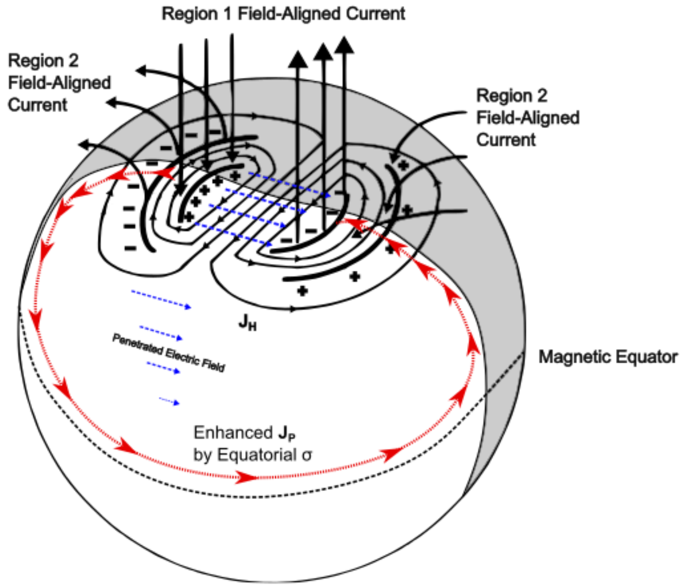


Figure 2: PPEF transmission circuit, from its propagation toward ionosphere through R1, to the horizontal transmission equatorward and the closure circuit through an equatorial Pedersen Current. Adapted from Wei et al. [2015]

rich air (enriched in $[O_2]$ and $[N_2]$), increasing the $[N_2]/[O]$ ratio. This shift in atmospheric composition enhances the loss rate of atomic oxygen ions (O^+) through recombination, which can lead to a negative ionospheric storm (TEC decrease) [Vijaya Lekshmi et al., 2011, Prölss, 1995].

Another potential effect of PPEF is their contribution to the intensity of geomagnetically induced currents (GIC), as these currents are driven by rapid changes in the geomagnetic field (dH/dt) over a region of interest, given the characteristic frequency of PPEF fluctuations. As Vishnupriya and Manju [2025] notes, PPEF can generate differences with respect to quiet electric field conditions at equatorial and low latitudes, particularly on the dayside. Conversely, Nilam and Tulasi Ram [2022] concluded that discrepancies between GIC models and dH/dt measurements from ground magnetometers could be explained by the omission of PPEFs and the equatorial electrojet, both of which are localized phenomena.

PPEF were studied in situ by Fejer et al. [1983], who observed their effects using incoherent scatter radar at Jicamarca. Subsequently, Le Huy and Amory-Mazaudier [2005] modeled for the first time the magnetic field associated with ionospheric disturbances, assuming that at mid-to-low latitudes these disturbances are primarily driven by PPEFs and disturbed dynamo electric fields (DDEFs). More recently, Younas et al. [2020] proposed the use of Fourier analysis to isolate magnetic fluctuations related to PPEFs based on their quasi-periodic behavior. Their approach employs high-pass filters for periods shorter than four hours and also considers the contribution of the asymmetry index (ASYH), thereby accounting for the magnetic effects of the PRC and FAC.

In our previous work [Castellanos-Velazco et al., 2024], we attempted to model ΔH and K_{loc} by applying the results from Le Huy and Amory-Mazaudier [2005] to filter the magnetic effects of prompt penetration electric fields (PPEFs) and disturbed dynamo electric fields (DDEFs) for 20 geomagnetic storms. Specifically, we added these filtered magnetic contributions to the Dst index after scaling it by a latitudinal factor. The objective was to assess whether the resulting Dst_λ time series exhibited reduced deviations from ΔH compared to the differences typically observed between Dst and ΔH. For certain events, we observed that the average difference between the expected Dst_λ and the measured ΔH was either smaller or larger than that between Dst and ΔH. Overall, the average reduction in difference was approximately 30%. Cases in which the reduction was negligible ($< 10\%$) likely correspond to events where the asymmetric ring current (ARC) significantly influenced the local response without being accounted for in the model. In contrast, events with a substantial reduction (around 60%) suggest that those geomagnetic storms exhibited a relatively weak contribution from the ARC.

For this study, we selected the geomagnetic storm of March 17, 2015, which was one of the events analyzed in our previous work [Castellanos-Velazco et al., 2024]. This storm is recognized as the most intense of solar cycle 24, reaching a minimum Dst of -274 nT during its main phase. An additional motivation for selecting this event is the availability of public magnetic data for the different regions of interest, as well as the relatively long duration of its MP (15 hours) which is crucial since the studied phenomena is strongly related to MP.

To analize the FAC activity related the ARC and how it influence the fluctuations related to PPEF, we are analyzing current density data provided by the Active Magnetosphere and Planetary Electrodynamics Response Experiment (AMPERE) [Anderson et al., 2020]. This program is carried on by satellite measurements with magnetometers which measure magnetic field over high latitudes. From ten minute recordings of magnetic field, they compute current density related to FAC coupling regions with high latitude ionosphere.

2 St. Patrick Geomagnetic Storm

The geomagnetic storm (GS) analyzed in this work is the intense event that occurred on March 17, 2015, widely known as the St. Patrick's Day storm. It is considered the most intense storm of solar cycle 24 Wu et al. [2016]. Where it was reported GNSS fails, scintillation and even possible affections due to GIC presence, according to Kozyreva et al. [2018].

The storm was triggered by a coronal mass ejection (CME) associated with a C9.1 solar flare and a subsequent dimming observed on March 15. This CME, recorded as a partial halo by the SOHO/LASCO coronagraphs, had an estimated speed of approximately 660 km/s.

Figure 3 provides an overview of the interplanetary conditions that led to the St. Patrick's Day storm. A more detailed description about the trigger event is provided by Wu et al. [2016]. Panel (a) shows the total interplanetary magnetic field, B_T (black line), and its north-south component, B_Z (red line). Panels (b) and (c) presents solar wind properties, including the induced Interplanetary Electric Field (E_Y), calculated as $-V_X \times B_Z$; and dynamic pressure. Finally, panel (e) displays the geomagnetic response through the SYMH and ASYH indices (green and yellow lines respectively).

The key milestones of the event are marked by three vertical dashed lines in Figure 3. The first line (left) indicates the arrival of an interplanetary shock (IPS) at 04:30 UT at the L1 point. The IPS signature is clear from the step-like increases in B_T, E_Y and P_{dyn}. This shock compressed the magnetosphere, causing a Sudden Storm Commencement (SSC) visible as a sharp increase in the SYMH index between 04:30 and 07:00 UT, highlighted by light gray shading.

The region following the shock, the CME sheath, is identified by highly fluctuating B_T and B_Z, enhanced P_{dyn}. A period of southward B_Z between 06:00 and 09:00 UT initiated the what we called, the first stage of main phase (MP), driving the SYMH index down to its first minimum of -80 nT.

The second dashed line (14:00 UT) marks the arrival of the magnetic cloud, identified by Wu et al. [2016] as the main driver of the GS. Its passage is identified by a smooth, strong rotation of the magnetic field, a drop in temperature, and so, a corresponding low plasma beta. A sustained and intense southward B_Z component (reaching -40 nT) within the cloud marked the beginning of the second (and more intense) stage of the MP. The SYMH index decreased sharply, reaching its global minimum of -234 nT at 22:47 UT. The entire main phase period is shaded in dark gray in panel (e).

The third dashed line (23:00 UT) indicates the trailing edge of the magnetic cloud and the entry into a high-speed solar wind (HSSW) stream. While the high solar wind speed persists, the geomagnetic activity subsides as B_Z turns northward and the E_Y drops close to zero, ending the geoeffective driving of the storm.

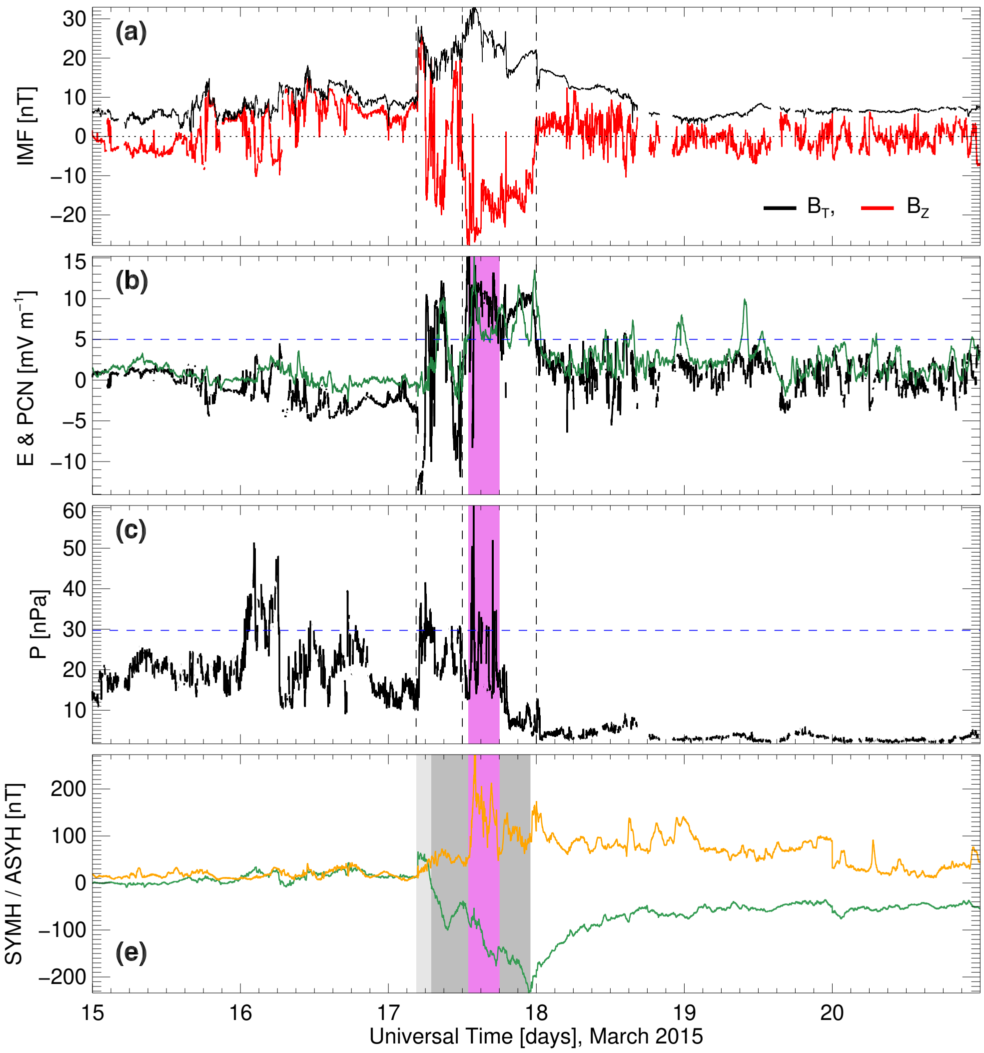


Figure 3: Interplanetary measurements. Panel (a): interplanetary magnetic field total component (black line) and z component (red line). Panel (b): E_Y Interplanetary Electric Field (blue line). Panel (c): Dynamyc Pressure [nPa]. Panel (d): SYMH (green line) and ASYMH index (orange line), highlighting the initial phase (light gray shade), main phase 1 (medium gray shade) and main phase 2 (dark gray shade).

Since we are interested in the effects of the PRC during its highest activity altogether with PPEF effects, we will focusing on a time window of a few hours during MP. This time window is going to be in function of periods when $E_Y > 0.5mVm^{-1}$ and $P_{dyn} > 29.71nPa$. The first criteria is based on the discussion in Burton et al. [1975], where they find the energy injection during MP is dependent of E_Y, setting $E_{Y0} = 0.5mVm^{-1}$ as a minimum value. On the other hand, Xie et al. [2008] set $P_{dyn} > 15nPa$ as a good estimator for a geoeffective dynamic pressure for strong geomagnetic storms. Considering that 15 nPa is a very low threshold for the period between march 15 - 17 as it can be seen in the Figure, we set the threshold through computing the 95% of Cumulative Density Function (CDF). Thus, we obtained a threshold of 29.71 nPa. Hence, obtain two possible time windows: 6-7 UT and 13-18 UT. Since the second period corresponds to the highest ASYH values, we selected this period for study, and its highlighted by the violet shading. Hence, during the selected time window, we expect higher injection and Electric Field penetration (under shielding) and PRC activity (Over shielding), which implies higher related magnetic fluctuations.

3 Procedure

3.1 Geomagnetic Data

Although more direct methods exist for studying prompt penetration electric fields (PPEFs)—such as incoherent scatter radar observations—these instruments are not widely distributed. The few available are typically located at either very high or equatorial latitudes, limiting their spatial coverage. In contrast, magnetometers offer a far more global distribution. Moreover, PPEF models, such as *(mention specific model here)*, tend to focus primarily on equatorial latitudes and may present certain limitations. For these reasons, analyzing magnetic fluctuations associated with PPEFs remains a valuable approach, particularly for regions at the boundary between mid and low latitudes.

The geomagnetic data analyzed in this paper were obtained from the INTERMAGNET public database (*International Real-Time Magnetic Observatory Network* [INTERMAGNET, 2021]), and from the Mexican Magnetic Service (SM) for the magnetic observatory (OB) located in Mexico. The selected OB are listed in Table 1, along with their respective geographic and geomagnetic coordinates. As shown in Table 1, the OB are situated within a magnetic latitude range of approximately $\lambda_m \sim 18°$ to $33°$. We focus on pairing OB with nearly opposite geomagnetic longitudes (ϕ_m) separated by $\phi_m \sim 180 - 200°$, expecting opposite local magnetic responses depending on their location. Analyzing how this opposing behavior varies across different pairs will help elucidate the influence of the ARC on the local response across different MLT sectors. In this context, MLT is used as a reference for the position of each OB relative to the subsolar point at noon.

Table 1: List of observatories considered for this study. Each OB is paired with another one at $\sim 180°$.

Observatory	IAGA code	Country/ Territory	Latitude λ Geographic	Latitude λ Magnetic	Longitude ϕ Geographic	Longitude ϕ Magnetic
Beijing	BMT	China	40.3 N	30.53 N	116.2 E	172 W
Guimar	GUI	Spain	28.317 N	33.21 N	16.433 W	61.1 E
Honolulu	HON	USA	21.317 N	21.49 N	158 W	90.03 W
Jaipur	JAI	India	26.917 N	18.42 N	75.8 E	150.47 E
Kakioka	KAK	Japan	36.232 N	27.8 N	140.186 E	149.73 W
San Juan	SJG	USA, Puerto Rico	18.11 N	28.19 N	66.15 W	6.1 E
Tamanraset	TAM	Argelia	22.783 N	24.3 N	5.571 E	82.29 E
Teoloyucan	TEO	Mexico	19.747 N	27.81 N	99.182 W	28.4 W

As mentioned previously, the analysis will cover the period of maximum energy injection, defined by a large interplanetary electric field component E_Y, high dynamic pressure, and peak asymmetric ring current (ARC) activity as measured by the ASYH index (reaching 273 nT). These conditions imply alternating phases of undershielding and overshielding of PPEF. Accordingly, we will focus on the first two days of the storm's development, specifically March 17 and 18, 2015

3.2 PPEF Magnetic fluctuations

It is well accepted that local recordings H_{loc} of a ground magnetometer can be considered a linear combination of different geomagnetic sources [Rishbeth and Garriott, 1969, Fukushima and Kamide, 1973, Knecht and B.M, 1985, Gjerloev, 2012, van de Kamp, 2013]. We show the main sources with the following Equation:

$$H_{loc} = H_{SQ} + H_0 + H_{MT} + H_{others} + H_I, \tag{1}$$

where H_{SQ} represents the diurnal variation and H_0 is the main field baseline for month window. H_{MT} represents the variations induced by the intensification of magnetospheric currents during the GS. In the case of H_{others}, it corresponds to magnetic sources less significant, like (and no limited to)) magnetotellurics, crustal magnetization, etc; which can be neglected, as for their magnitudes and variations are not considered as relevant compared to the others mentioned. Finally, H_I Le Huy and Amory-Mazaudier [2005] corresponds to the magnetic variations driven by ionospheric current activity during the GS.

The local geomagnetic response during a GS is given by:

$$\Delta H = H_{loc} - (H_{SQ} + H_0), \tag{2}$$

The quiet-time baseline H_{SQ} was computed by adapting the method of van de Kamp [2013], which involves applying a low-pass filter to extract the first six harmonics from the average of the five quietest days. The baseline H_0, on the other hand, was obtained using a least squares fitting procedure applied to undisturbed days, with each data point derived from the average of nighttime measurements.

From Le Huy and Amory-Mazaudier [2005], for mid-low magnetic latitudes, with λ as the geomagnetic latitude of each OB, we can roughly approximate H_{MT} as $H_{MT} \approx SYMH \cdot \cos(\lambda)$. Thus, H_I can be approximated as:

$$H_I = H_{DDEF} + H_{PPEF} + H_{others} \approx H - (H_{SQ} + H_0 + SYM - H \cdot \cos(\lambda)), \tag{3}$$

where H_{DDEF} and H_{PPEF} represent the magnetic fields associated with the disturbed dynamo electric fields and the prompt penetration electric fields, respectively. H_{others} on the other hand remains as a noise.

In order to isolate H_{PPEF} from H_I, we proceed under the assumption that both types of ionospheric electric fields drive quasi-periodic fluctuations [Nishida, 1968a]. Accordingly, we apply a Fourier-based band-pass filter for frequencies between 69.5 and $550, \mu$Hz, corresponding to periods ranging from 4 hours to 30 minutes [Younas et al., 2020]. A limitation of applying Fourier analysis directly to Equation 3 is that it only accounts for the symmetric component of the ring current, neglecting the asymmetric part. As shown by Crooker and McPherron [1972], Fukushima and Kamide [1973], Iyemori [1990], the magnetic contribution of the asymmetric ring current (ARC) can be comparable to—and during certain periods of the main phase even exceed—that of the symmetric ring current. Consequently, a substantial portion of H_I may arise from the ARC rather than from PPEFs and DDEFs. To address this, Younas et al. [2020] proposes incorporating an additional term that accounts for the asymmetric ring current prior to filtering, using the ASYH index as shown in the following equation::

$$H_I' = H_I - ASYH \cdot cos(\lambda). \tag{4}$$

According to Iyemori [1990], the ASYH index represents the asymmetry of the ring current during a geomagnetic storm. Meanwhile, Fukushima and Kamide [1973] concludes that, at mid-to-low latitudes, FAC—and by extension the PRC—constitute the primary contribution to this asymmetry. Therefore, it is reasonable to expect that most of the residual effects on H_I from Equation 3 may be associated with FAC activity. However, given that the asymmetric ring current intensifies and decays more rapidly than the symmetric RC [Fukushima and Kamide, 1973], and considering the complexity of the underlying system, Equation 4 should be applied with certain considerations.

Hence, in order to understand how the asymmetry of RC would affect H_I, we analyze FAC activity at the MLT sectors corresponding with the OB points during the periods of high asymmetry. Hence, we analyze how H_I' is affected by removing the effects of the ACR by Equation 4, in comparison with H_I once we filter H_{PPEF}.

4 Results

In the following section, we focus on the results related to the geomagnetic response of Equation 3 during march 17 and 18 using the data from the ground magnetometers listed in Table 1. Then we compare the response of H_I in function of MLT during high asymmetry periods, with FAC activity Jr.

Figure 4 presents an overview of the magnetic H_I response compared to the asymmetric activity of RC (ASYH index), which is at the top panel. The magnetic contribution H_I from ionosphere phenomena is displayed in magenta and green lines in panels (b) through (e), corresponding to observatories with a geomagnetic longitude difference of approximately $\Delta\phi_m \sim 150 - 180°$. In the bottom panel (e), H_I is

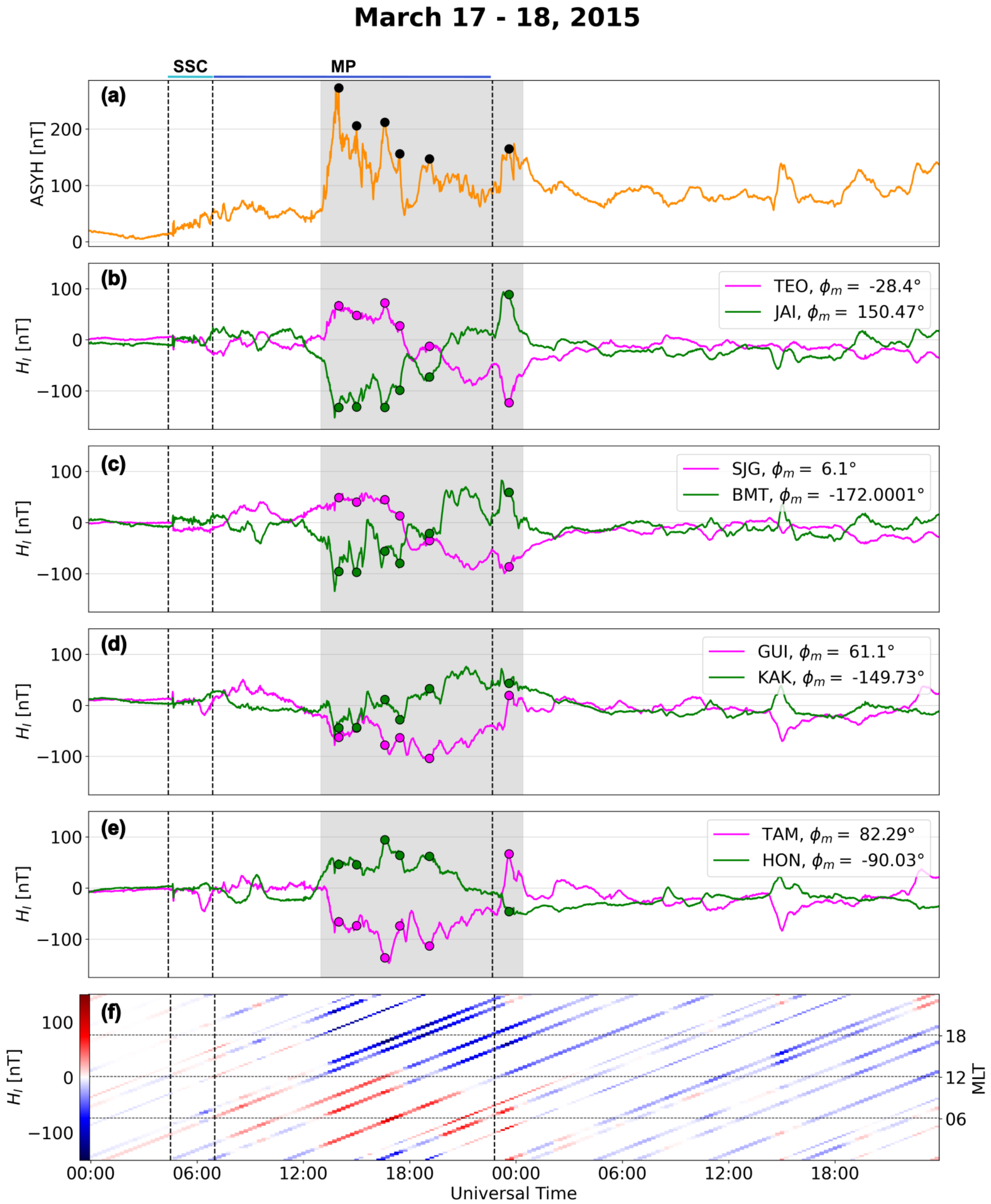


Figure 4: Comparison between ASYH index (top panel) with H_I from Eq 3, taking in each panel, OB at $\Delta\phi_m \sim 180°$. The colored points represent periods of high asymmetry within RC system and how each H_I. The last panel is a mapping of H_I with respect to MLT and UT. The dashed vertical lines represent the beginning of SSC, MP and RP.

mapped as a function of MLT over time in UT. The dashed vertical lines indicate the transitions between the quiet period and the sudden storm commencement (SSC), the SSC and the MP and RP. The gray shading highlights the period of highest RC asymmetry, as captured by the ASYH index.

Notice how each OB pair exhibits opposing behavior in their respective H_I signals, as expected given their $\Delta\phi_m$ separation. By selecting specific points, represented in the Figure 4 by filled circles during periods of high asymmetry, distinct trends can be highlighted with an apparent variation as we observe more eastern/western OB.

Considering 14 UT as the reference time until 2 UT form next day (shaded area), the behavior of H_I corresponds to ascending and descending responses, for OB with initial MLT positions in morning–afternoon (TEO and SJG) or afternoon–morning (JAI and BMT) transitions. In these cases, the local response exhibits nearly linear intensification or attenuation over time. Another case occur when the OB point is initially located near noon or midnight (panel d and e). At these panels, H_I begin with low magnitudes, to grow and gradually attenuate at the end. If we take H_I within the shaded area in longitudinal sequence from TEO to TAM, then from JAI to HON, it seems to follow a sinusoidal pattern, which makes sense considering the geometry and rotation of the earth. It also tells about how ARC affects H_I differently at global scale.

The regions where H_I is mostly influenced by the ARC can be more easily tracked in panel (f), where positive and negative H_I values are represented using red and blue color scales, respectively, similar to what Iyemori [1990] did. The distribution of positive and negative H_I is sharply defined during the period of highest ASYH activity. Specifically, positive H_I values are most prominent around 3–9 MLT, while negative H_I dominates in the 15–21 MLT sector. This clear contrast in polarity persists until approximately 02 UT on March 18, a few hours after the end of the MP, as indicated by the SYMH minimum. Beyond this point, positive H_I values are no longer sustained, appearing only briefly during minor injection events. Additionally, a gradual migration of the positive H_I response can be observed, shifting from around the noon sector toward earlier MLT.

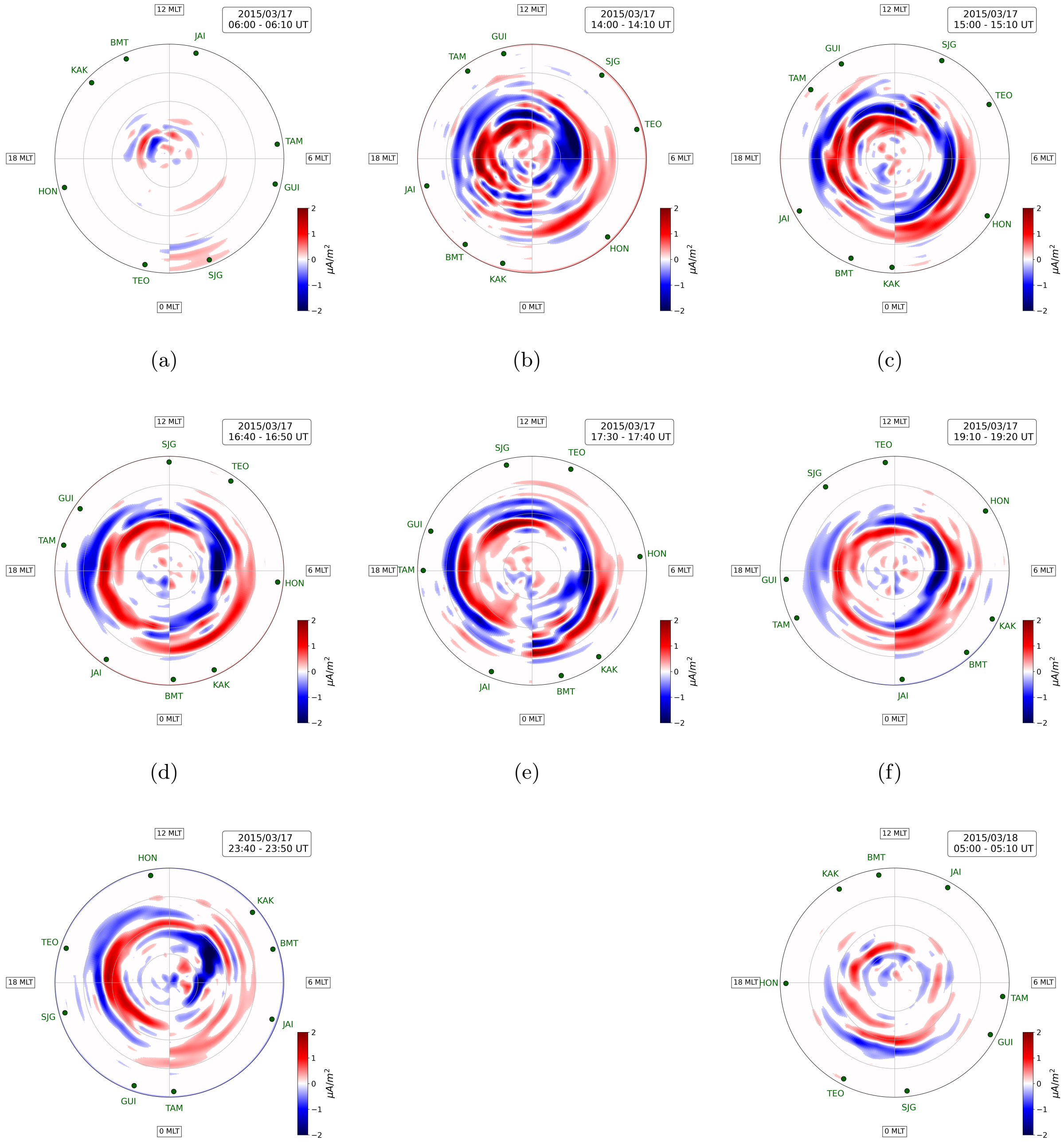
Once we found that H_I at mid-low latitudes follows a MLT distribution, similar to what we observe on the FAC distribution at high latitude ionospheric regions, the next step is comparing it when the OB was at certain MLT positions during FAC activity. In other words, we compared the radial current density $\mathbf{J_r}$ from AMPERE data with H_I in the same MLT sector during periods of high RC asymmetry.

Figure 4 presents eight time windows during periods of high asymmetry, as identified in Figure 4. These correspond to periods of high ASYH during following times: 14:07, 15:07, 16:43, 17:33, 19:14, and 23:43 UT (panels b-g). Each window displays the current density $\mathbf{J_r}$ with a radial direction (into and out of the page) using a red–blue color scale, where red denotes upward (positive) and blue downward (negative) currents. Panels (a) and (h) show reference periods before and after the main energy injection into the magnetosphere, specifically at 06:00 UT on March 17 and 05:00 UT on March 18, respectively. In each panel, the MLT positions of the observatories are indicated by green labeled dots.

In panels (b) through (g), a high concentration of ascending and descending radial current density $\mathbf{J_r}$ is observed within the 70°–80° magnetic latitude range, corresponding to the R1-FAC coupling regions with the ionosphere. Three main current regimes can be identified: a predominantly ascending current region ($\mathbf{J_r} > 0, \mu\text{A/m}^2$) centered around 18–19 MLT, a predominantly descending current region ($\mathbf{J_r} < 0, \mu\text{A/m}^2$) centered around 6–7 MLT, and transition regions near noon and midnight.

Being consistent with panel (f) of Figure 4, we observe in panel (b) that TEO and SJG are located in a MLT sector with mostly descending currents. There, the color scale indicates $\mathbf{J_r} < -2, \mu\text{A/m}^2$ at R1-FAC latitudes, contrasting with $\mathbf{J_r} \sim 1, \mu\text{A/m}^2$ in the R2-FAC region and a much weaker gradient. A similar pattern is observed for TAM and JAI in the same panel, which coincide with ascending currents ($\mathbf{J_r} > 2, \mu\text{A/m}^2$) and a steep gradient relative to the downward R2-FAC on the dusk side. Analogously, stronger H_I responses would be expected for GUI and HON in panel (c); HON in panel (d); and KAK, BMT, TEO, and SJG in panel (e), based on their alignment with these FAC-dominant regions during periods of high asymmetry.

Given that FAC constitute the primary source of the ARC [Fukushima and Kamide, 1973], we compute



(a) (b) (c)

(d) (e) (f)

(g) (h)

Figure 5: Radial Current Density, parallel to the geomagnetic field, provided by AMPERE. Blue represents downward current density meanwhile red represents the upward current. Panels (b–g) correspond to different times of high asymmetry, meanwhile (a) and (h) panels represent periods before and after the high injection activity period.

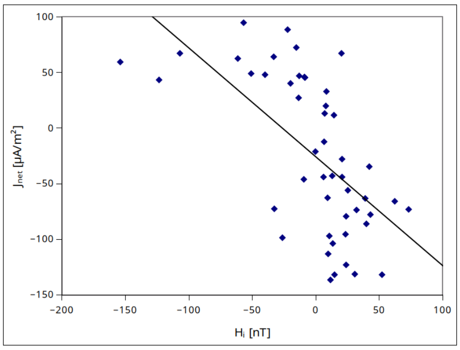


Figure 6: Correlation Analysis ASYH and H_I, for pairs OB longitudinally separated $170-200°$. Red circle marks values with more contribution from PRC

the net current density J_{net} integrated over a specific MLT coinciding to each OB position. Considering a simple projection from high latitudes to mid-low latitudes, we seek to confirm whether the J_{net} is ascending or descending, in comparison with H_I response at the previously mentioned times. Figure 6 presents a scatter plot of J_{net}, versus H_I for the six sampled periods previously mentioned. The computed correlation coefficient is $\rho = -0.61$. This anticorrelation suggests that energy injection from R1-FAC translates into a positive H_I response in the 5–9 MLT sector, while the strongest negative H_I response is associated with the 15–19 MLT sector, being consistent with H_I distribution seen in Figure 4. These results further support the significant influence of FAC but also, how this influence on Equation 3 may evolve through time for each OB.

The next step is analyzing how FAC possibly affected the resulting reconstruction of magnetic fluctuations related to PPEF. We show in Figure 7, both: H_{PPEF} after filtering results of Eq 3 and H'_{PPEF} after filtering results of Eq 4.

In general, H'_{PPEF} exhibits higher amplitudes than H_{PPEF}. However, for TEO, SJG, KAK, and HON, H_{PPEF} and H'_{PPEF} appear to show a phase shift during the interval from 12 to 19 UT. Interestingly, the other four observatories display lower H'_{PPEF} amplitudes compared to H_{PPEF} for fluctuations around 0 UT. What each group of four observatories has in common is their initial MLT position at 14 UT, the onset of the highest injection period. These groupings are consistent with the MLT positions shown in Figure 5b: HON, TEO, and SJG were located in the region of predominantly descending current, while TAM, JAI, and BMT were situated in areas of predominantly ascending current. GUI and KAK, on the other hand, were initially positioned near the transition regions.

5 Discussion

To achieve a reasonable reconstruction of the magnetic fluctuations associated with prompt penetration electric fields (H_{PPEF}), it is necessary, as previously mentioned, to subtract as possible the magnetic contributions from the sources outlined in Equation 1. For the magnetospheric source, we use the SYMH index multiplied by a factor of $\cos(\lambda_m)$, where λ_m is the magnetic latitude of the location of interest, as an approximation of the magnetospheric contribution. This approach, proposed by Le Huy and Amory-Mazaudier [2005] and expressed in Equation 3, primarily accounts for RC activity. This is reasonable, since most of the disturbance recorded by ground magnetometers at mid-to-low magnetic latitudes during a geomagnetic storm is indeed attributable to the RC [Fukushima and Kamide, 1973]. However, this formulation overlooks the contribution of the asymmetric component of the RC system. As a result, this asymmetric part remains within H_I and constitutes the majority of its magnitude, as seen in 4, where during different periods, H_I from different OB may follow a similar or an opposed pattern.

The way of how H_I trend evolves as we go toward eastern OB, so as how H_I presents positive and negative responses in function of MLT reinforces the idea of regions of higher or lower ARC influence over H_I. Considering this, we decided to compare H_I responses during high asymmetry with the FAC activity within ionosphere at high latitudes. We made a comparison between H_I at mid-low latitudes

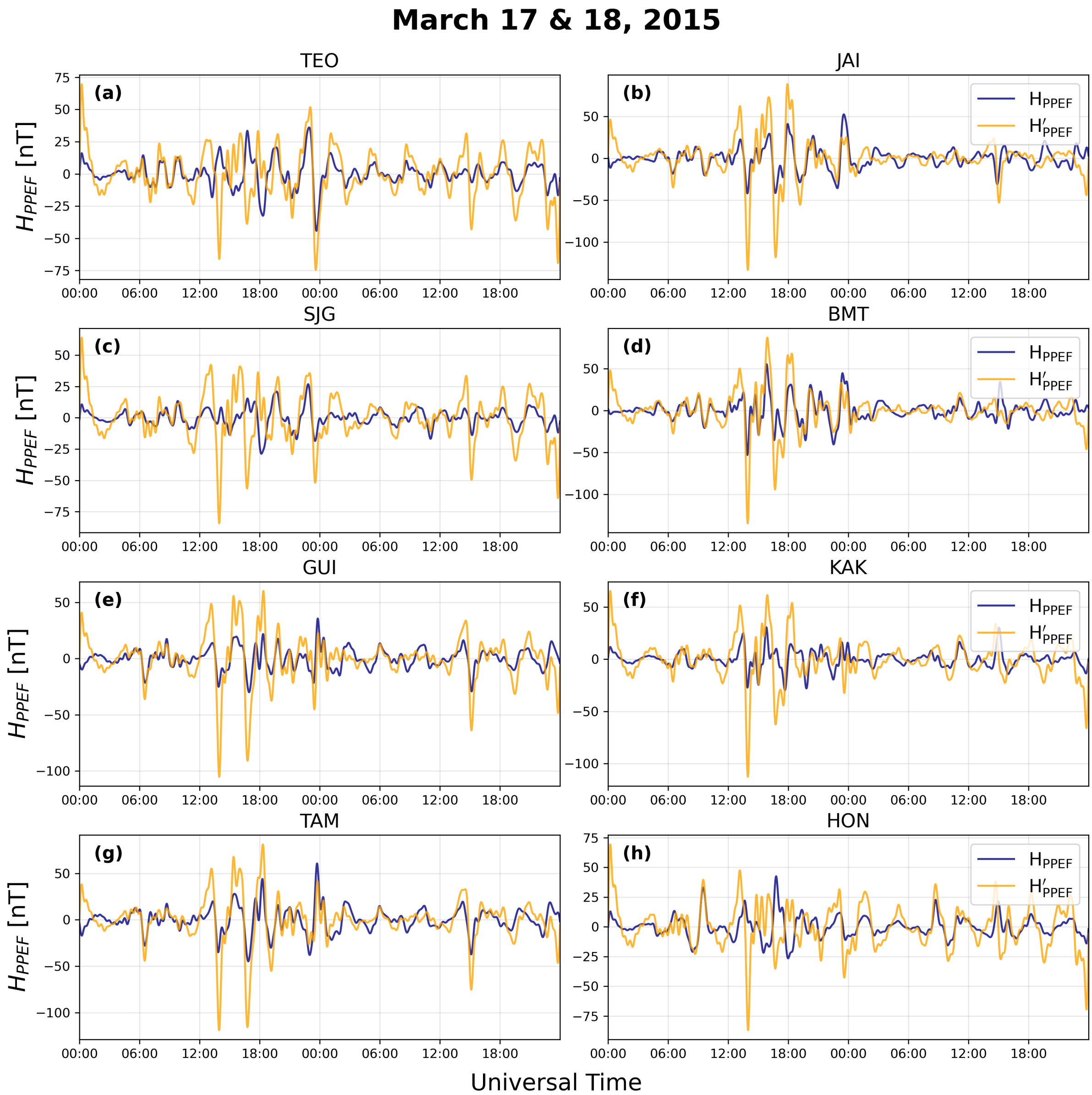


Figure 7: H_{PPEF} reconstruction from Eq 3(blue lines) and from Eq 4, after removing asymmetric RC contribution (orange line).

with $\mathbf{J_r}$ at high latitudes from considering that conservation of $\mathbf{J_r}$ requires a three-dimensional closed circuit extending through the magnetosphere. This complete loop—comprising FAC, the PRC, and high-latitude ionospheric closure currents—generates magnetic fields at mid and low latitudes. According to Fukushima and Kamide [1973], this is why FAC constitute the primary source of the ARC at mid-to-low magnetic latitudes. Nevertheless, it must be considered that the observed effect at these latitudes is dominated by the contribution of the curved FAC themselves, and is modulated by coupling and shielding processes within the ionosphere. This represents a limitation of this first-order approach.

The trend observed in the scatter plot of Figure 6, which compares J_{net} with H_I, is consistent with the findings of Iyemori [1990], who noted that the ASYH index exhibits a strong correlation with injection periods and, consequently, with R1-FAC activity. Therefore, it is reasonable to infer that during the selected periods, there is a higher contribution from R1-FAC, with $J_{net} > 0$ (predominantly downward current) around the dawn sector and $J_{net} < 0$ (predominantly upward current) around the dusk sector. This is confirmed with the highest J_{net} seen from TEO at 14:07 UT ($J_{net} = -106\mu\text{A/m}^2$) so as BMT ($J_{net} = -153\mu\text{A/m}^2$) and KAK ($J_{net} = -123\mu\text{A/m}^2$) at 23:40 UT .

The correlation coefficient $\rho = -0.61$ indicates that H_I response tends to be negative or positive, and consequently ΔH intensified or attenuated, depending on whether a given observatory is located in MLT sectors around dusk or dawn during periods of peak energy injection. This is consistent with the fact that FAC activity constitutes the primary contribution to ring current asymmetry [Fukushima and Kamide, 1973]. However, the correlation is not particularly strong. One of the reasons for this discrepancy may be attributed to the PRC, which, according to Kamide and Fukushima [1971], Crooker and McPherron [1972], contributes to negative H_I values primarily in the 19–23 MLT sector during the early stages of the main phase, thereby enhancing the negative H_I response beyond what would be expected from FAC activity alone.

As illustrated in Figure 1 for the northern hemisphere and, in the equatorial plane perspective, the connection regions can be downward (FAC–PRC) on the eastern side, and upward (PRC–FAC) on the western side, near the dusk sector. The PRC and more specifically, its western side most likely coincides with the 19-23 MLT range. By focusing on OB situated within this range during the sampled periods, we find results consistent with this interpretation. For example, in panel (b), at the TAM position (14 MLT), a net current density of $J_{net} = 62, \mu\text{A/m}^2$ coincides with $H_I = -70, \text{nT}$, while JAI at 19 MLT, $J_{net} = 52\mu\text{A/m}^2$ corresponds to $H_I = -140\text{nT}$. In the same period, BMT at 21 MLT coincides with $J_{net} = 62\mu\text{A/m}^2$ showing $H_I = -140\text{nT}$. Similarly, in panel (c), GUI and TAM exhibit $J_{net} = 43$ and $11, \mu\text{A/m}^2$, respectively, while their H_I values are -76 and -135nT. Comparable patterns are observed in the other sampled periods for observatories located within the 16–20 MLT range.

The case of the eastern region of the PRC is way less notorious, several examples illustrate the complexity of the relationship between J_{net} and H_I. In panel (c), both BMT and KAK exhibit upward J_{net}, yet their H_I responses differ markedly: BMT shows $H_I = -59, \text{nT}$, while KAK—located farther east—registers $H_I = 11, \text{nT}$, despite both observatories being situated at relatively close geomagnetic longitudes. Similarly, in panel (d), BMT's location presents a J_{net} close to zero with an associated $H_I = -20, \text{nT}$, whereas at JAI, $J_{net} = 10, \mu\text{A/m}^2$ corresponds to a much more negative $H_I = -114, \text{nT}$. These contrasting examples highlight the influence of localized PRC dynamics and the limitations of using J_{net} alone to estimate the asymmetric magnetic response at mid-latitudes.

From what we have seen, ARC influence is complex to be considered by just taking $ASYH \cdot cos(\lambda_m)$ from H_I. This turns out to be important when evaluating the reconstruction of magnetic fluctuations associated with PPEF using both approaches (Equations 3 and 4). As shown in Figure 7, there are cases during certain periods where H'_{PPEF} exhibits significant amplification of its fluctuations compared to H_{PPEF}, as well as instances where, rather than amplification, a phase shift in the fluctuations becomes apparent.

Hence we analyzed the distribution of the magnetic fluctuations (Figure 8). The distribution of H_{PPEF} is represented by blue bars meanwhile H'_{PPEF} is represented by yellow bars. In both cases, they approximate to normal distributions. Accordingly, we fitted Gaussian models for each case: solid red lines correspond to the H_{PPEF} model, and dashed red lines to the H'_{PPEF} model. The estimated

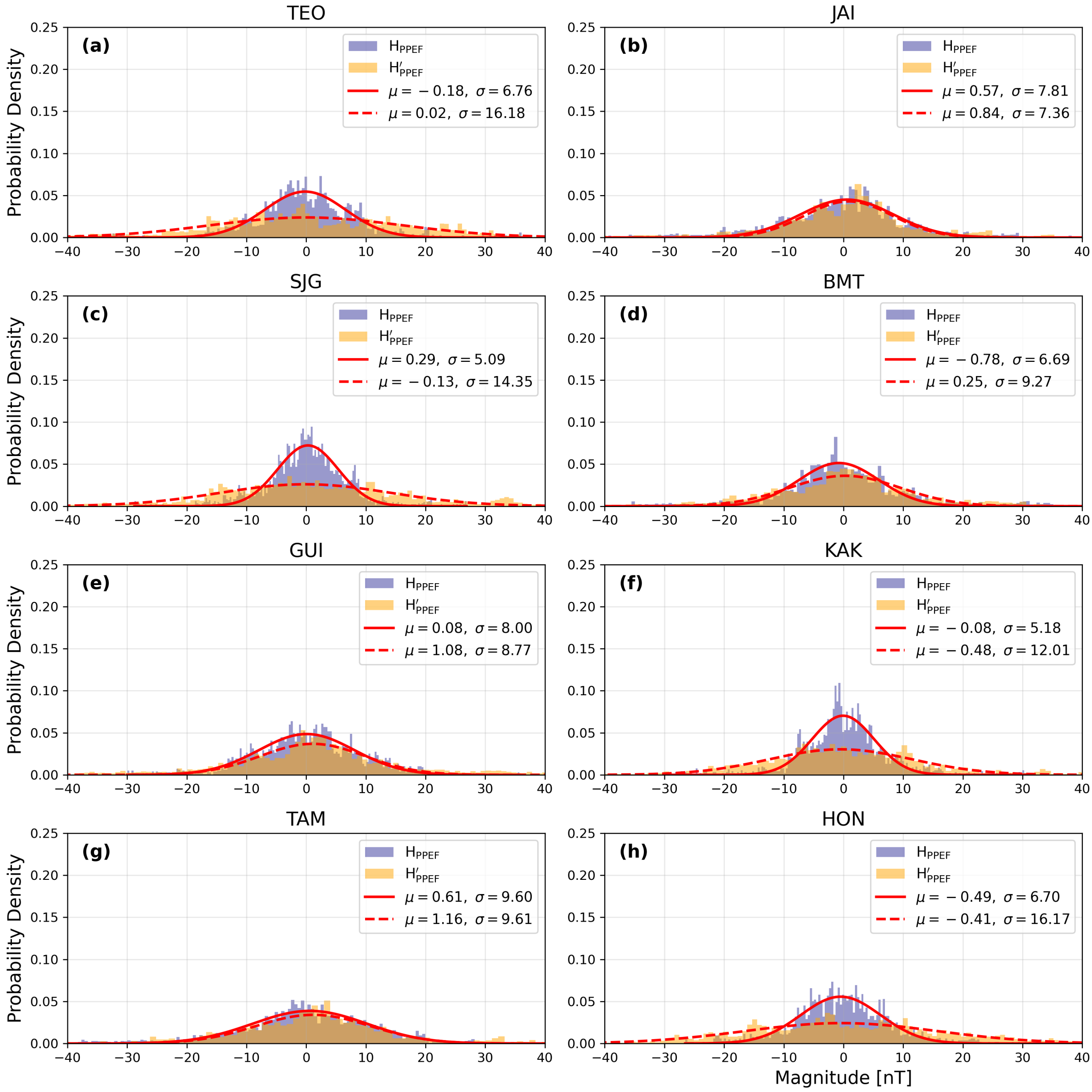


Figure 8: H_{PPEF} distribution for each OB. Color dashed vertical lines represent the a threshold for H_{PPEF} to be considered as intense amplitudes

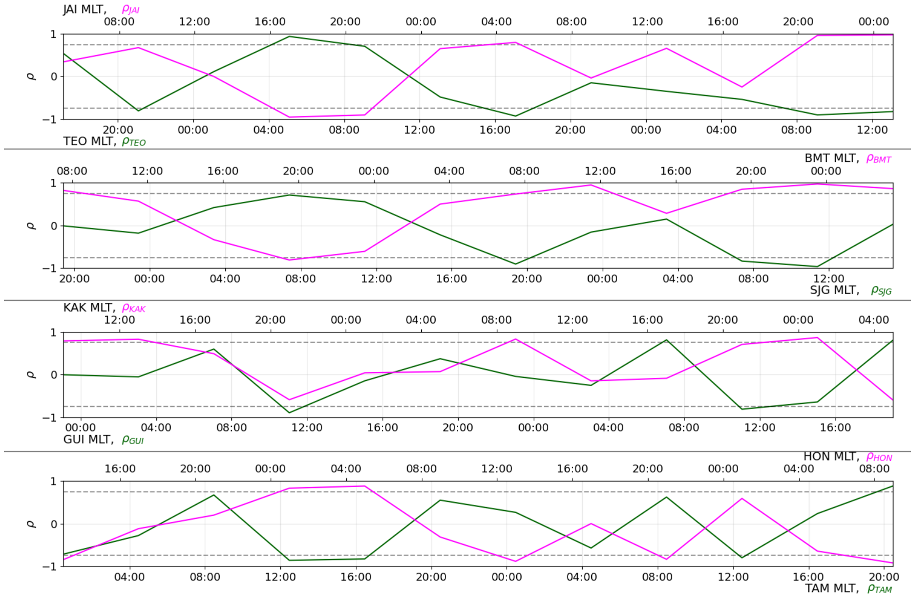


Figure 9: Correlation Analysis ASYH vs H_I through 4 hours periods for pairs OB longitudinally separated $170 - 200°$. Each X-axis is labeled by their respective MLT series.

median (μ) and standard deviation (σ) for each fitted model are displayed in the corresponding panels.

It was found that H'_{PPEF} may exhibit minor, significant, or pronounced increases in the dispersion (σ) of the data depending on the observatory. These differences appear to be related to the initial MLT position of each OB at the moment of maximum injection (14 UT). For GUI, TAM, and JAI—all located in the afternoon to early evening sector, as shown in Figure 5b—the fitted distribution parameters show only minor changes between the H_{PPEF} and H'_{PPEF} models. In contrast, TEO, SJG, HON, and KAK, which lie within the night to morning sector, exhibit a clear enhancement in dispersion, reflected by a notable increase in σ. BMT, whose initial position appears to correspond to a transition region, presents an intermediate behavior consistent with its location.

Hence, it is clear that FAC can affect H_I differently depending on the initial MLT position of a given OB. Returning to the question of how reliable it would be to apply Equation 4 in a general manner, we observe in Figure 4 that, for some observatories, H_I may follow either a similar or opposite pattern to the ASYH index during certain periods.

Given the previously discussed chain of influence, from FACs to the ACR, and from FACs to H_I, it is reasonable to further examine the correlation between ASYH and H_I. We found that most correlation coefficients lie within the range of -0.5 to 0.5. Although these values would not typically be considered indicative of significant (anti)correlation, this result is consistent with the fact that the ARC affects H_I differently depending on the MLT position of each observatory during the event, as also evident in Figure 4. This implies that high positive or negative correlation coefficients ($\pm\rho$) would only be expected during relatively short intervals. To account for this, we applied a moving correlation window of four-hour bins, as shown in Figure 9, with dashed lines indicating thresholds at $\rho = \pm 0.75$, which we consider as strong (anti)correlation.

Consistent with our earlier observations, there are periods of both strong and weak (anti)correlation. Three of the four observatory pairs exhibit a high possible influence of ASYH on H_I during the period of maximum asymmetry (12–16 UT on March 17). Later, the pairs in panels (a) and (b) again show strong (anti)correlation after a 4–6 hour interval, which aligns with these observatories reaching the afternoon or morning sectors. However, these tendencies are not always observed. For example, in the case of GUI and KAK, which were located closer to the transition sectors (near noon and midnight) during the peak injection period, the correlation remains weak.

The variability in results—both across different observatories for the same event and even for a single observatory across different geomagnetic storms, as shown in Castellanos-Velazco et al. [2024], demonstrates that the ARC does not always interfere in the same way. The longitudinal extent of the influence of different FAC regions, as well as that driven by the PRC, varies not only spatially but also over time. Therefore, Equation 4, introduced by Younas et al. [2020], must be applied with caution, taking into account the specific position of the observation point during the period of highest energy injection. Method proposed by Younas et al. [2021] wouldn't be an option either since it is mostly focused on anti-correlating H_I from diurnal variations in order to obtain H_{DDEF}.

6 Conclusions

Within this project, we analyzed the magnetic fluctuations driven by prompt penetration electric fields (PPEFs) during the intense geomagnetic storm (GS) of March 17, 2015. To this end, we used geomagnetic data from four pairs of ground-based observatories separated by approximately 160° to 200° in magnetic longitude. Additionally, we incorporated data from the AMPERE program to examine field-aligned current (FAC) activity during the period of interest, focusing on the same magnetic local time sectors corresponding to the locations of the ground magnetometers along the evolution of the GS.

We applied the different approaches proposed by Le Huy and Amory-Mazaudier [2005] and Younas et al. [2020] to study the influence of the asymmetric ring current (ARC) system on the magnetic signature associated with prompt penetration electric fields (H_{PPEF}). We also sought to compare the net current density, considering its predominant upward or downward direction, in order to assess the degree of influence on H_I over the specific magnetic local time (MLT) sectors where the observatories are located.

Regarding the end of the main phase (MP), the delimitation point is not as clear-cut as it might seem. Depending on the initial MLT position of an observatory, it may continue to experience effects associated with the main phase for several hours after the reconnection process has ceased, as observed in Figure 4. Naturally, this behavior depends on the intensity of the geomagnetic storm and the rate of energy injection during the MP.

During the analysis, we attempted to approximate the degree of influence of the ARC on the residual magnetic field H_I, obtained after removing the diurnal variation, the main field, and the magnetospheric contributions. After comparing the results with the ASYH index and the activity of FAC, we arrived at the following conclusions:

- Even though the contribution of PRC to the ARC is not as significant as the net contribution from FAC, it can still induce important local differences in H_I, particularly on its western side.
- Depending on the initial position of the observation point, the influence of the ARC will vary not only from the onset but throughout the development of the MP and the evolution of the RC system.
- The higher responses of H_I related to ARC can be tracked to the afternoon-evening (most negative) side and the morning side (most positive). In contrast, around noon and midnight sectors, ARC influence might not so relevant.
- The residual influence of the ARC can significantly affect H_{PPEF} at the moment of being filtered from H_I, particularly in terms of its amplitude and phase fluctuations.

Due to all the considerations discussed above, it is necessary to adopt an approach that accounts not only for the ARC system but also—where possible—for the relative weight it exerts over a specific point or region of interest, as well as the time intervals during which this influence is most relevant. Such an approach would improve the modeling of H_{PPEF} and could also enhance the estimation of geomagnetically induced currents (GICs) over that particular region.

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